FARMPULSE AI · ARCHITECTURE PROPOSAL B

FarmPulse Cloud

A hybrid edge–cloud, event-driven platform for multimodal animal monitoring, digital profiles and early-risk alerts across many farms

VERSION	AUDIENCE	STATUS
1.0 · 24 October 2026	FarmPulse AI team, supervisors, future partners	Proposal for review

The idea in one line

Keep only real-time perception at the farm on a small edge gateway, stream compact events to a cloud platform, and build identity, behaviour, audio, profiles, risk fusion, decisions and notifications as independent services around an event bus.

Who it is for

The full vision in the project proposal: cameras plus microphones plus farm and vet records, a persistent digital profile per animal, multimodal temporal fusion, herd analytics, digital passport and marketplace, for many farms.

Why this shape

Models improve centrally from every farm's data, each student work package becomes a separately deployable service, compute scales with demand, and adding a farm means plugging in a gateway rather than building another server.

What it costs

More moving parts (Kubernetes, Kafka, MLOps), roughly $1,100–2,100 a month of cloud spend during the pilot, and a longer path to the first end-to-end demo. Health features depend on connectivity; security alerts do not.

1 Context and scope
2 Architecture drivers
3 Architecture overview
4 Components
5 Data flow and event model
6 AI, ML and MLOps
7 Technology stack
8 Trade-offs and decisions
9 Risks and mitigations
10 Scalability and performance
11 Security, privacy and multi-tenancy
12 Cost estimate
13 Rollout plan
14 Evaluation and success criteria
A Mapping to the existing code
B Comparison with Proposal A

1 Context and scope

The project proposal (*FarmPulse_AI_Final.pdf*) frames FarmPulse AI as a livestock platform that continuously understands each animal using cameras, microphones, historical records and farm data. Instead of only detecting an animal in a frame, it builds a long-term **digital profile** for every animal and flags meaningful changes in behaviour, activity, feeding, drinking, growth and vocalisation. It is explicitly a decision-support and early-warning system, not a diagnostic one. The proposal splits the AI work across four students (detection and re-ID; behaviour; vocalisation; multimodal temporal fusion) and lists farm operations, a digital passport and a marketplace as connected modules.

The project README keeps the same pipeline but stages it: vision first, then audio, with a suspicion classifier, RL prioritisation and alerts over dashboard, SMS and WhatsApp. Proposal B designs the **target platform** that the whole vision needs, while sequencing the rollout so vision-only value arrives first.

> Core research question (from the proposal): can continuous visual behaviour and animal vocalisation analysis, combined with each animal's historical baseline, provide useful early warnings of abnormal livestock conditions and help farmers identify animals that require inspection?

Starting point

The folder already holds a working video prototype (`cow-detection`: YOLO11 detection, cattle-tuned BoT-SORT, ID re-linking, zone behaviours, FastAPI dashboard), a ready-to-train Siamese hide-pattern identification library (`cow_hide_identification`: ResNet-50, 128-d embeddings, triplet loss), farm footage from handheld phones and a fixed CCTV camera, and 13 reference papers. These cover AI 1 and part of AI 2 in offline form. Nothing yet runs continuously, persists across days, or connects to alerts.

In scope for Proposal B

- Multiple farms (design target 50, pilot 1–3), each with cameras and later microphones, and multiple user roles: owner, farm worker, veterinarian serving several farms, administrator, buyer.
- All four AI work packages as services: identity and tracking, behaviour, audio, multimodal temporal fusion and risk; plus the README's suspicion window and RL prioritisation.
- Animal digital profile (identity, breed, age, weight and growth, vaccinations, medication, vet records, activity timeline, audio history, risk alerts, marketplace status).
- Farm operations (medication and vaccination schedules, vet visits, inventory, missed-treatment alerts), weekly and herd analytics.
- Digital passport with verifiable records and a marketplace listing that links to it.
- A continuous MLOps loop: sampling, labelling, training, evaluation, staged rollout to edge and cloud.

Out of scope

- Diagnosis or automated treatment; payments and logistics in the marketplace (listings and verification only); species other than cattle and buffalo in the first versions, as the proposal recommends.

2 Architecture drivers

Driver	Requirement	How Proposal B responds
Multi-farm scale	Add farms without new servers or code forks; vets and buyers see across farms.	Multi-tenant cloud, farm_id on every event and row, thin standard gateway per farm.

Bandwidth	Raw streams (about 4 Mbps per 4 MP camera) cannot be uploaded from rural farms.	Edge does detection, tracking and embeddings; only events, keyframes and short tracklet clips go up (about 0.5–1 Mbps average per farm).
Multimodality	Video, audio and records must be aligned per animal and time.	Common event envelope and time base; a profile service that fuses streams; a feature store with per-animal baselines.
Model improvement	Models should learn from all farms and be redeployed safely.	Central data lake, labelling, registry, shadow and canary rollout, signed OTA updates to gateways.
Team structure	Four students own four AI problems with separate datasets and metrics.	One service per work package with a published event contract; independent release cycles.
Latency	Intruder alerts in seconds; health alerts within minutes.	Security handled on the edge; health chain targets 10 minutes from evidence to notification.
Trust	Passports and marketplace need tamper-evident records; README requires separation of measured, predicted and confirmed data.	Event-sourced history, signed passport documents, separate outcome store, full lineage of model versions.

3 Architecture overview

Proposal B is a **hybrid edge–cloud, event-driven microservice platform**. Each farm runs an edge gateway that turns camera (and later microphone) streams into compact, identity-ready events. Those events travel over MQTT with mutual TLS to a cloud broker and onto a Kafka event bus. Domain services consume and publish events, each owning its own data. Users reach the platform through an API gateway from a web dashboard, a farmer progressive web app and a vet portal. A separate MLOps plane closes the loop from production data back to better models.

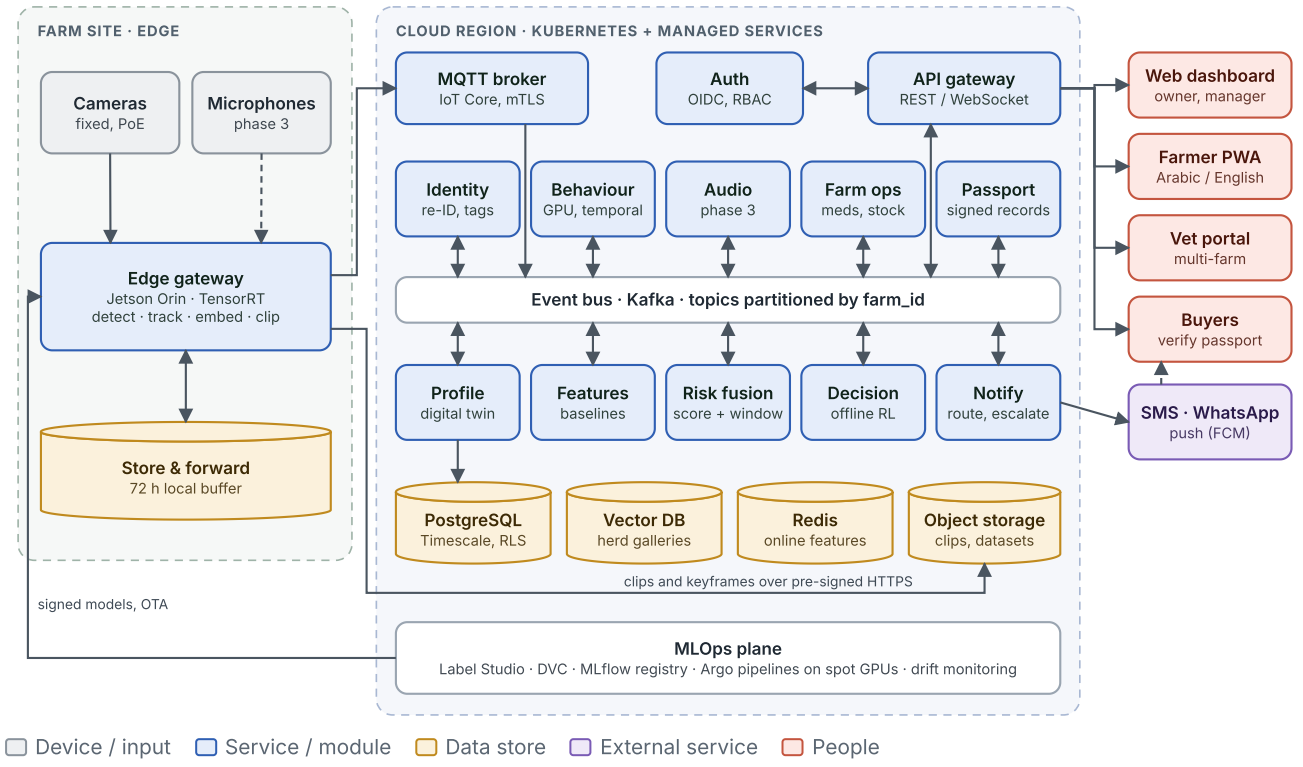


Figure 1. Proposal B component view. Gateways publish events over MQTT (mTLS) and upload clips through pre-signed URLs. Every cloud service publishes and consumes on the event bus and owns its own store.

Key architectural choices

- **Edge for perception, cloud for understanding.** Frame-rate work (detection, tracking, embedding, clip cutting, intruder rules) stays local. Anything that benefits from history, many farms or large models (re-ID against galleries, temporal behaviour models, audio, fusion, decisions) runs in the cloud.
- **Events as the contract.** Services integrate only through versioned events (CloudEvents envelope, schema registry). This lets the four AI services evolve independently and lets the system replay history when a model improves.
- **Event-sourced animal profile.** The digital profile is a projection of immutable events (observations, treatments, weights, outcomes). Passports are signed snapshots of that history, so a buyer can verify where each fact came from.
- **Safety rails around learning.** The risk model and the RL decision service always run behind rules, confidence thresholds and human approval. New models go through shadow mode and canary rollout before they can change what a farmer sees.

4 Components

Edge (per farm)

Component	Responsibility	Technology
Edge gateway	Decode RTSP, run cow detection and tracking at 5–10 fps, compute 128-d hide embeddings per tracklet, cut 2–4 s tracklet clips and keyframes, apply the intruder rule locally, publish events, and expose health telemetry. One Jetson Orin NX 16 GB per 4–6 cameras.	Ultralytics YOLO11 or DeepStream with TensorRT, BoT-SORT, `cowid` embedder, GStreamer
Store & forward	Buffer events and clips for up to 72 h when the uplink drops, then replay in order with idempotent event IDs.	SQLite / local disk queue
Local safety path	Intruder alerts trigger a local siren or relay and go out through the gateway's cellular modem even when the cloud is unreachable.	GPIO / relay, SMS modem
Fleet agent	Receive signed model and software updates, report versions, roll back on failure.	AWS IoT Greengrass v2 or balena

Cloud services

Service	Responsibility	Owns	Package
Identity	Resolve tracklets to registered animals using the farm's embedding gallery, ear-tag OCR results and camera topology for cross-camera re-ID; keep unassigned when unsure; route doubtful matches to a review queue; learn new animals.	Galleries (vector DB), identity links	AI 1
Behaviour	Classify behaviour on tracklet clips (eating, drinking, lying, standing, walking, rumination, grazing) with a temporal model, combined with zone rules from the edge; output behaviour intervals with confidence.	Behaviour intervals	AI 2
Audio (phase 3)	Denoise and segment microphone audio, classify vocalisations with CNN + BiGRU + attention, keep zone-level attribution unless one animal can be justified, combine with visible mouth-open events.	Audio events	AI 3
Profile & records	Maintain the digital profile per animal from events; one record per animal per 5-minute window; registration, breed, age, weight, treatments, vet notes.	Profiles, observations (Timescale)	Platform
Features	Compute hourly and daily features and personal baselines; serve them online for scoring and offline for training with point-in-time correctness.	Feature store (Redis + Parquet)	AI 4
Risk fusion	Multimodal temporal model over vision, audio and records; outputs a low, medium or high risk score, an estimated time window and the main contributing signals.	Risk scores	AI 4
Decision	Prioritise risky animals and suggest an approved action; fixed policy first, then a contextual bandit or offline RL policy evaluated off-policy and run in shadow mode.	Decisions, policy logs	AI 4
Notify	Route by priority and role, escalate if unacknowledged, respect quiet hours, record delivery and acknowledgement; SMS, WhatsApp templates, web push.	Alerts, deliveries	Platform

Farm ops	Medication and vaccination schedules, vet check-in and check-out, inventory of feed and medicine, missed-treatment alerts; later feed-demand forecasting.	Ops records	Platform
Passport & market	Issue signed passport documents from profile history (Ed25519 signatures, hash-chained record digests), public verification page, marketplace listings referencing passports.	Passports, listings	Extension
API gateway, Auth	Single entry for apps; OIDC login, roles, tenant scoping, rate limits; WebSocket for live alerts.	Users, roles	Platform

5 Data flow and event model

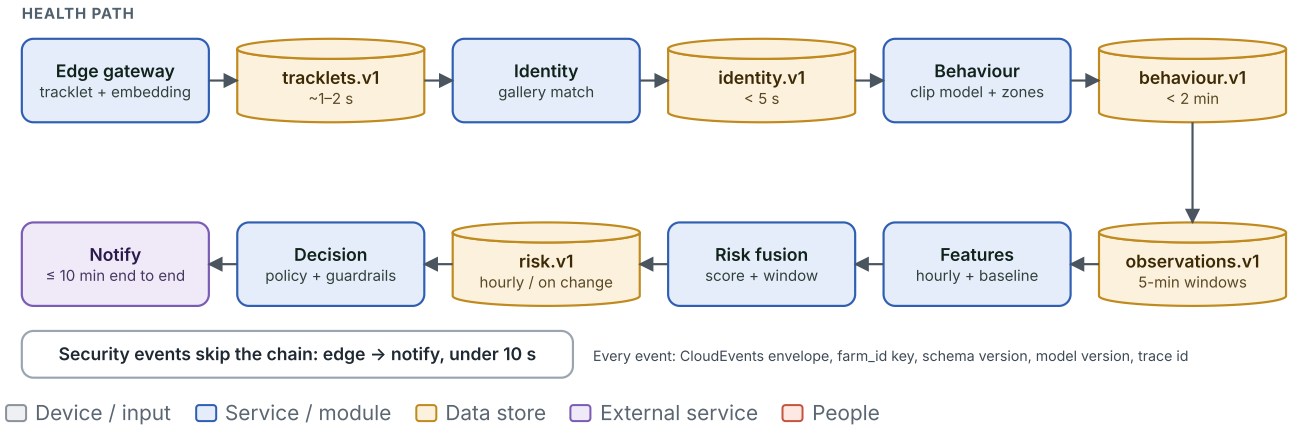


Figure 2. Health path through the event bus with target latencies. Topics are versioned; consumers can replay a topic when a model or rule changes.

1. **Edge perception.** The gateway closes a tracklet (or emits a heartbeat every 30 s for long ones) with box statistics, zone dwell times, the hide embedding, keyframe and clip references. The clip goes to object storage through a pre-signed URL; the event goes to MQTT.
2. **Identity.** The identity service matches the embedding against the farm's gallery, merges in tag OCR when present, applies camera-topology rules, and publishes `identity.v1` with method and score, or *unassigned*.
3. **Behaviour.** The behaviour service scores the tracklet clip with the temporal model on a GPU pool autoscaled by topic lag, merges zone evidence and publishes behaviour intervals.
4. **Records.** The profile service folds identity and behaviour (and audio in phase 3) into one observation per animal per 5-minute window.
5. **Features and risk.** Features update hourly and on significant change; the risk service scores animals whose features moved and publishes risk with a time window and explanation.
6. **Decision and notify.** The decision service assigns priority and action; notify sends to the right people and channels and tracks acknowledgement.
7. **Feedback.** Vets and owners confirm, reject and record outcomes in the app. Outcomes go to their own topic and store, then into training sets and policy evaluation.

Event envelope (example)

```
{
  "specversion": "1.0", "type": "farmpulse.identity.v1", "id": "01JD…", "source": "svc/identity@2.3.1",
  "time": "2026-11-02T06:07:12Z", "farmid": "farm-07", "traceparent": "00-4bf9…",
  "data": {
    "tracklet_id": "farm-07/cam-03/2026-11-02T06:06:40Z/17", "animal_id": "C104",
    "method": "gallery+tag", "score": 0.93, "alternatives": [["C221", 0.61]],
    "model": {"embedder": "cowid-resnet50-128@1.2.0", "gallery_version": 41}
  }
}
```

Data stores

Store	Contents	Notes
PostgreSQL + TimescaleDB	Profiles, observations, features history, risk, decisions, alerts, outcomes, ops records, passports	Managed (RDS or Cloud SQL), row-level security by farm, compression after 30 days

Vector DB	Per-farm embedding galleries by view	pgvector to start, Qdrant if galleries grow large
Redis	Online features, alert de-duplication keys, sessions	Managed, no durable data
Object storage	Clips, keyframes, audio segments, datasets, model artefacts, Parquet lake	Lifecycle rules: clips 30 days, labelled data kept
Kafka	All topics, 7-day retention; compacted topics for latest state	Partitioned by farm_id for ordering per farm

6 AI, ML and MLOps

Package	Model design	Data
AI 1	Edge: YOLO11 fine-tuned on farm footage, BoT-SORT, embedding per tracklet. Cloud: metric-learning re-ID (current Siamese ResNet-50 to start, then self-supervised multi-camera training as in MultiCamCows2024), tag OCR, and a detect-segment-identify path for dense crowds (2602.15962). Solid-coloured animals rely on tags.	Farm data, OpenCows2020, MultiCamCows2024
AI 2	Temporal model on tracklet clips: start with X3D or VideoMAE fine-tuned on CBVD-5, then a Cattle-CLIP style video-language model for few-shot new behaviours; zone and posture rules stay as features and as a fallback.	CBVD-5, CattleBehaviours6, XGain, farm clips
AI 3	Noise reduction and segmentation, log-mel spectrogram, CNN + BiGRU + attention from scratch as in the proposal, with a pretrained audio encoder as a comparison; outputs normal or abnormal vocalisation and event classes.	CowRumIA, Daylong Cow Acoustic, farm recordings
AI 4	Risk fusion: a temporal model (GRU with attention or Temporal Fusion Transformer) over hourly features and deviations from each animal's baseline, with discrete-time survival heads for 24, 48 and 72 h; calibrated probabilities; explanations per alert. Starts as a rule set. Decision: fixed policy → contextual bandit → conservative offline RL (IQL or CQL), with off-policy evaluation and shadow mode.	Outputs of AI 1–3, vet outcomes, ops records

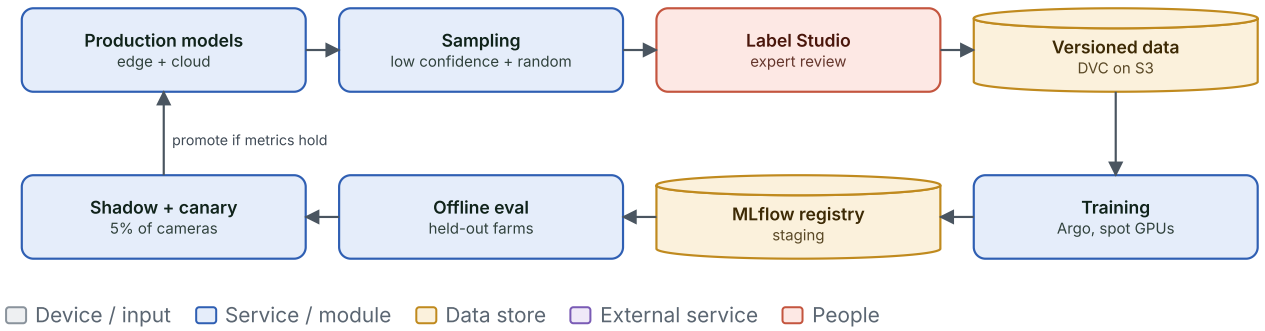


Figure 3. Continuous learning loop. No model reaches farmers without offline evaluation on held-out farms and a shadow or canary period.

- **Sampling.** Low-confidence detections and identities, disagreements between rule and model, alert-related clips and a random sample go to labelling.
- **Labelling.** Label Studio with per-task guidelines; a vet reviews health labels; inter-annotator agreement tracked.
- **Training.** Argo Workflows on spot GPU nodes; datasets pinned with DVC; every run logged to MLflow with data and code versions.

- **Evaluation.** Splits by farm, animal and period; per-condition metrics (night, crowding, breed); fairness across coat types.
- **Release.** Registry stages (staging, shadow, canary, production); edge models compiled to TensorRT per device type, signed, rolled out to 5% of cameras first, with automatic rollback on metric drop.
- **Monitoring.** Drift on input statistics and confidence; alert precision from feedback; weekly model report.

7 Technology stack

AWS is used as the reference cloud because its IoT and managed Kafka services fit well; every choice has an open-source or other-cloud equivalent (in brackets) to avoid lock-in.

Layer	Choice
Edge	NVIDIA Jetson Orin NX 16 GB, JetPack, TensorRT, Ultralytics or DeepStream, GStreamer, Greengrass v2 (balena)
Ingestion	AWS IoT Core MQTT with X.509 device certificates (EMQX or Mosquitto), pre-signed S3 uploads
Event bus	Amazon MSK Serverless (Redpanda or Apache Kafka on Kubernetes), schema registry with Protobuf or Avro
Compute	Amazon EKS (any Kubernetes), Karpenter for nodes, KEDA autoscaling on topic lag, GPU node pool with L4 GPUs on spot
Services	Python 3.12, FastAPI, aiokafka, Pydantic; Triton Inference Server for GPU models
Data	RDS PostgreSQL 16 + TimescaleDB-compatible extension or self-managed Timescale, pgvector (Qdrant), ElastiCache Redis, S3 (MinIO), Parquet + Athena (Trino)
ML	PyTorch, Ultralytics, torchaudio, LightGBM, PyTorch Forecasting, d3rlpy, Feast, MLflow, DVC, Label Studio, Argo Workflows, Evidently
Front ends	Next.js web dashboard and vet portal; installable PWA for farmers with Arabic and English and offline views
Identity	Amazon Cognito (Keycloak), OIDC, role- and tenant-based access
Messaging	Twilio SMS and WhatsApp or Meta WhatsApp Cloud API, Firebase Cloud Messaging for push
Platform	Terraform, Argo CD (GitOps), GitHub Actions, OpenTelemetry, Prometheus, Grafana, Loki, Tempo, AWS KMS and Secrets Manager (Vault)

8 Trade-offs and decisions

Decision	We gain	We give up
Split perception (edge) from understanding (cloud)	Low bandwidth, central learning, large models where they are cheap to run	Two runtimes to build and test; health features pause during long outages (buffered, not lost)
Microservices around Kafka	Independent services per student, replayable history, natural fan-out to analytics, passport and marketplace	Operational complexity, eventual consistency, harder local debugging
Managed cloud services	Less undifferentiated operations work, built-in durability	Monthly cost from day one, some vendor coupling (mitigated by open equivalents)
Event-sourced profile	Full audit trail, tamper-evident passports, rebuild projections when logic changes	More design effort; schema evolution needs discipline
Central MLOps with canary rollout	Safe, measurable model improvement across farms	Labelling and pipeline effort before the first model update
Multi-tenant from the start	Vet portal and marketplace work across farms; cheaper per farm at scale	Tenant isolation must be right everywhere; higher stakes for security bugs

9 Risks and mitigations

Risk	Level	Mitigation
Platform work crowds out the research (Kubernetes, Kafka, IaC take weeks)	High	Start with a "walking skeleton" on Docker Compose with Redpanda locally; use managed services; one person owns platform; cut scope per phase.
Too few verified outcomes for risk fusion and RL	High	Rules as baseline; structured outcome capture in the vet portal; report retrospective and shadow results only.
Cloud cost overruns (GPUs left on, verbose logs, clip uploads)	Medium	Budgets and alerts, spot and scale-to-zero GPUs, upload quotas per farm, log sampling, monthly cost review.
Rural connectivity gaps delay health insights	Medium	72 h store and forward, local security path, events small enough for 2G/3G fallback.
Detection and identity errors in crowds, at night and on solid-black animals	High	Farm-specific fine-tuning, tag OCR at chokepoints, review queue, per-condition metrics, active learning on failures.
Tenant data leak or compromised gateway	Medium	Row-level security, per-device certificates and topic ACLs, penetration test before multi-farm pilot.
Microphone attribution is ambiguous in shared barns	Medium	Zone-level audio events unless attribution is justified; fusion treats audio as herd-zone context.
Marketplace trust and legal obligations	Low	Listings only, passports signed and verifiable, legal review before any transaction features.

10 Scalability and performance

50 farms	< 10 s	≤ 10 min	~3M rows/day
design target, pilot 1–3	intruder alert (edge path)	evidence to health alert	observations at 50 farms × 200 animals

- **Edge.** Scale by adding gateways: one Orin NX per 4–6 cameras at 5–10 fps (to benchmark). Gateways are stateless apart from the buffer and can be swapped.
- **Ingestion.** 50 farms × 8 cameras × about 2 events per second is roughly 800 events per second, small for Kafka. Partition by farm_id (for example 24 partitions) to keep per-farm ordering and allow parallel consumers.
- **GPU services.** Behaviour and audio inference run on Triton with dynamic batching on a GPU node pool; KEDA scales replicas by topic lag and to zero at night where clips are sparse.
- **Storage.** 50 × 200 animals × 288 windows a day is about 2.9 million observation rows a day; Timescale compression keeps this in the tens of GB per year. Clips are lifecycle-managed in object storage.
- **Bandwidth per farm.** Example: 200 animals × 6 clips per hour × 3 s × 1 Mbps (cropped 720p) ≈ 1 Mbps average plus small events; tunable per farm.
- **Availability.** Multi-AZ database and Kafka; services stateless with at least two replicas; target 99.5% for the health path, with the edge path independent of the cloud.

11 Security, privacy and multi-tenancy

Area	Controls
Devices	Per-gateway X.509 certificate in a secure element where available; MQTT topic ACLs limited to the gateway's own farm; secure boot and disk encryption on Jetson; signed OTA artefacts; cameras on an isolated VLAN with defaults changed.
Network	TLS 1.2+ everywhere, mTLS between gateway and broker, private subnets for services and databases, WAF in front of the API gateway, no public database endpoints.
Identity and access	OIDC with MFA for staff and vets; roles (owner, worker, vet, admin, buyer); tenant claim in every token; PostgreSQL row-level security on farm_id; vets granted per farm with expiry.
Data protection	KMS encryption at rest for databases, buckets and Kafka; secrets in Secrets Manager; data residency by choosing the closest suitable region; retention rules per data class; person blurring on stored clips.
Integrity and audit	Immutable event log; audit trail for outcome edits and passport issuance; passports signed with keys held in KMS; public verification endpoint.
Supply chain	Container image scanning, SBOMs, pinned dependencies, signed images (cosign), least-privilege IAM roles per service.
Responsible AI	Decision support only, approved action list, human approval, explanations shown with every alert, shadow mode for policies, documented limitations per model.

12 Cost estimate

Indicative 2026 list prices in USD (AWS on-demand unless noted). Real bills depend on region, discounts, usage and messaging rates; treat these as planning ranges.

Edge hardware per farm	Range
Jetson Orin NX 16 GB kit, enclosure (per 4–6 cameras)	$900–1,300
4–8 IP cameras (or reuse CCTV)	$300–900
Microphones + audio interface (phase 3)	$150–400
PoE switch, cabling, UPS, 4G router	$400–700
Total per farm	**≈ $1,750–3,300**

Cloud, pilot of 1–3 farms, per month	Range
Kubernetes control plane + 3 general nodes	$220–300
GPU inference (1 L4 GPU, spot / scaled)	$200–600
Managed Kafka (or Redpanda on nodes)	$0–400
Managed PostgreSQL (Multi-AZ) + Redis	$280–400
Object storage, transfer, IoT messaging	$30–100
Observability, secrets, DNS, WAF	$60–150
SMS, WhatsApp, push	$10–60
Total	**≈ $1,100–2,100**

Training on spot GPUs adds roughly $100–300 a month during active research. Much of the cloud bill is fixed platform cost, so the cost per farm falls with scale: at 50 farms the estimate is roughly $40–90 per farm per month. For a student pilot, cloud credits (education and start-up programmes) and running Redpanda and Postgres on the cluster can bring the pilot bill toward the bottom of the range.

13 Rollout plan

Phase	Weeks	Work	Exit criterion
0 · Walking skeleton	1–6	Event schemas and envelope; Terraform for a dev environment; local Docker Compose with Redpanda, Postgres and MinIO; auth; CI/CD; one gateway publishing fake events end to end to a dashboard tile.	Event travels gateway → bus → service → app
1 · Vision on one farm (M1–M2)	4–14	Gateway with fine-tuned detector, tracker and embedder; identity service with gallery and review queue; behaviour service with zone rules and first temporal model; labelling pipeline; data contract with the vet.	Held-out metrics for AI 1 and AI 2
2 · Records, risk v0, alerts (M3)	12–20	Profile service and observations; features and baselines; rule-based risk with time window; fixed decision policy; notify with SMS, WhatsApp and push; farmer PWA and vet portal; weekly reports.	Live alert on all channels, one traced example to outcome
3 · Audio and fusion	18–28	Microphones at the farm; audio service; multimodal risk fusion model trained and evaluated against rules; MLOps loop with shadow and canary.	Fusion beats single-modality baselines on held-out periods
4 · Decisions and operations (M4)	24–32	Contextual bandit or offline RL policy in shadow; farm ops module (vaccinations, inventory, vet visits); passport issuance and verification.	Policy evaluation report; first signed passports

5 · Multi-farm pilot	after 32	Second and third farm; tenant isolation tests; cost and reliability review; marketplace listings.	Three farms live within budget

The same team of four can map to services: AI 1 owns gateway perception and identity, AI 2 behaviour, AI 3 audio (and helps with platform until phase 3), AI 4 features, risk and decisions. Platform and front-end work is shared and is the main schedule risk.

14 Evaluation and success criteria

Area	Metric	Initial target (to confirm)
Identity	Top-1 and Recall@5 across cameras and days; share of observations assigned; review-queue volume	Top-1 ≥ 90% on patterned cows across days
Behaviour	Per-class F1 on held-out farm clips; minutes-per-day error	Feeding and lying time within ±15%
Audio	Per-class F1 and abnormal-event recall on held-out recordings	Set after dataset review
Risk fusion	Recall of vet-confirmed cases, false alerts per 100 animal-days, lead time; gain over best single modality	Agreed with the vet before testing
Decision	Off-policy value versus fixed policy; vet agreement	Non-inferior before any live use
Platform	End-to-end latency, event loss, availability, cost per farm	p95 ≤ 10 min, zero loss, ≥ 99.5%

A Appendix · Mapping to the existing code

Existing file	Role in Proposal B
`detector.py`, `tracker.py`, `trackers/*.yaml`	Core of the edge gateway perception loop, exported to TensorRT for Jetson.
`relink.py`	Split between edge (within-camera fragment linking) and identity service (cross-time linking with galleries).
`behaviour.py`, `zones.py`, `posture.py`	Zone dwell features computed on the edge; rules kept in the behaviour service as features and fallback.
`video_processor.py`, `app.py`, `static/`	Become a developer tool for offline re-processing and labelling review; production UI moves to the web dashboard.
`cow_hide_identification/src/cowid`	Embedder runs on the edge; training moves into the MLOps pipeline; galleries live in the vector DB.

B Appendix · Comparison with Proposal A

Dimension	Proposal A · FarmPulse Edge	Proposal B · FarmPulse Cloud
Shape	Modular monolith on one GPU server per farm	Edge gateways + cloud microservices on an event bus
Best fit	Current README stage: vision, one farm, M1–M4	Full proposal: multimodal, many farms, passport, marketplace
Offline behaviour	Everything keeps working; alerts via GSM	Security works; health insights wait for the uplink (buffered)

Time to first end-to-end demo	About 12–20 weeks	About 18–24 weeks
Cost	≈ $2,100–3,600 per farm once, ≈ $20–70 a month	≈ $1,750–3,300 per farm once, ≈ $1,100–2,100 a month for the pilot
Scaling	Vertical, one box per farm	Horizontal, shared platform
Main risk	Single node and limited growth	Platform complexity and cost for a small team

Prepared from the project proposal, both README versions, the code in `cow-detection` and `cow_hide_identification`, recorded run summaries and the 13 reference papers in the project folder. Performance and cost figures are estimates to be confirmed during the milestones.